



Lab Activity

Week 7: Introduction to Lists - Collections and Operations

Duration: 55 minutes — Work in pairs!

Name: _____ Partner: _____

Lab Goals

By the end of this lab, you will be able to:

- Create and modify lists using indexing
- Add and remove items with `append()`, `insert()`, `remove()`, and `pop()`
- Search lists using `in`, `index()`, and `count()`
- Build lists dynamically with loops
- Apply common list patterns (accumulator, find, minmax)

Getting Started

1. Extract the given [week7_lab.zip](#) file to your desktop
2. Remember: Lists are MUTABLE - we can change them!
3. Indexing starts at 0, not 1!

1 Exercise 1: List Basics and Indexing (8 minutes)

Get comfortable with creating lists and accessing elements!

Open [week7_ex1.py](#) and complete:

```
1  # Create your first lists
2  scores = [85, 92, 78, 88, 95]
3  names = ["Ali", "Sara", "Ahmed", "Fatima"]
4  mixed = ["Python", 3, True, 9.5]
5
6  # Access elements - fill in the indices
7  print("First score:", scores[___])      # Should print 85
8  print("Third score:", scores[___])      # Should print 78
9  print("Last score:", scores[___])      # Use negative index
10
11 # Modify elements
12 scores[1] = ___ # Change second score to 100
13 print("Updated scores:", scores)
14
15 # List operations
16 print("\nList info:")
17 print("Length:", len(scores))
18 print("Maximum:", max(scores))
19 print("Minimum:", min(scores))
```

```
20 print("Sum:", sum(scores))
21 print("Average:", sum(scores) / len(scores))
```

Practice Questions:

1. What is `names[-2]`? _____
2. What happens if you try `scores[10]`? Try it! _____

✔ **Checkpoint:** Can you access and modify list elements? Show teacher!

2 Exercise 2: List Methods - Adding Items (9 minutes)

Master the methods for adding items to lists!

Open [week7_ex2.py](#) and complete:

```
1  # Start with empty shopping list
2  shopping = []
3  print("Initial list:", shopping)
4
5  # append() - add to end
6  shopping._____("milk")
7  shopping._____("bread")
8  shopping._____("eggs")
9  print("After append:", shopping)
10
11 # insert() - add at specific position
12 shopping._____(__, "butter") # Insert at index 1
13 print("After insert:", shopping)
14
15 # Add multiple items at once
16 more_items = ["cheese", "yogurt"]
17 shopping._____(more_items) # Use extend() method
18 print("After extend:", shopping)
19
20 # Practice: Build a todo list
21 print("\n=== TODO LIST ===")
22 todos = []
23 # Add 3 tasks using append
24 todos._____("Finish homework")
25 todos._____(__) # Add your own task
26 todos._____(__) # Add another task
27
28 # Insert urgent task at beginning
29 todos._____(__, "Study for test")
30
31 print("Your todos:", todos)
32 print(f"You have {len(todos)} tasks!")
```

💡 **Tip:** `append()` adds to end, `insert()` adds at any position!

👥 **Pair Programming:** Switch who types every 5 minutes!

3 Exercise 3: List Methods - Removing Items (9 minutes)

Learn different ways to remove items from lists!

Open [week7_ex3.py](#) and complete:

```

1  # Fruit inventory
2  fruits = ["apple", "banana", "orange", "banana", "grape", "banana"]
3  print("Original:", fruits)
4
5  # remove() - removes first occurrence by value
6  fruits.remove("banana") # Remove first banana
7  print("After remove:", fruits)
8
9  # pop() - removes by index and returns it
10 removed = fruits.pop() # Remove last item
11 print(f"Popped '{removed}'")
12 print("After pop:", fruits)
13
14 # pop with index
15 first_fruit = fruits.pop(0) # Remove first item
16 print(f"Removed '{first_fruit}' from start")
17 print("After pop(0):", fruits)
18
19 # Check before removing
20 print("\n== Safe Removal ==")
21 grades = [85, 92, 78, 88, 95]
22 remove_grade = 78
23
24 if remove_grade in grades: # Check if exists
25     grades.remove(remove_grade)
26     print(f"Removed {remove_grade}")
27 else:
28     print(f"{remove_grade} not found!")
29
30 print("Final grades:", grades)

```

Challenge: Remove ALL occurrences of a value

```

1  numbers = [1, 2, 3, 2, 4, 2, 5]
2  # How to remove all 2's? Think about it!
3  # Hint: Use a while loop

```

✓ **Checkpoint:** Can you add and remove items correctly?

4 Exercise 4: Searching and Counting (9 minutes)

Find what you need in your lists!

Open [week7_ex4.py](#) and complete:

```

1  # Game high scores
2  scores = [1500, 2300, 1800, 2300, 1200, 2500, 2300]
3
4  # Check if value exists

```

```

5 target = 1800
6 if target ___ scores:
7     print(f" Score {target} found!")
8
9     # Find its position
10    position = scores._____(target)
11    print(f"   At position: {position}")
12 else:
13     print(f" Score {target} not found")
14
15 # Count occurrences
16 high_score = 2300
17 count = scores._____(high_score)
18 print(f"\nScore {high_score} appears {count} times")
19
20 # Practice: Student name search
21 print("\n=== CLASS ROSTER ===")
22 students = ["Ali", "Sara", "Ahmed", "Fatima", "Ali"]
23
24 search_name = input("Search for student: ")
25 if search_name ___ students:
26     # Find first position
27     first_pos = students._____(_____)
28     # Count total
29     total = students._____(_____)
30
31     print(f" Found {search_name}")
32     print(f"   First occurrence: position {first_pos}")
33     print(f"   Total count: {total}")
34 else:
35     print(f" {search_name} not in class")

```

 **Tip:** Always check with 'in' before using index() to avoid errors!

5 Exercise 5: Building Lists with Loops (10 minutes)

Create lists dynamically through user input and calculations!

Open [week7_ex5.py](#) and complete:

```

1 # Pattern 1: Collect numbers from user
2 print("=== NUMBER COLLECTOR ===")
3 numbers = [] # Start empty
4
5 count = int(input("How many numbers? "))
6 for i in range(count):
7     num = int(input(f"Enter number {i+1}: "))
8     numbers._____(_____) # Add to list
9
10 print("\nYour numbers:", numbers)
11 print("Sum:", sum(numbers))
12 print("Average:", sum(numbers) / len(numbers))
13
14 # Pattern 2: Generate list with calculation
15 print("\n=== SQUARE NUMBERS ===")
16 squares = []
17 for num in range(1, 11):
18     squares._____(num ** 2) # Add square of number

```

```

19
20 print("Squares 1-10:", squares)
21
22 # Pattern 3: Filter and build new list
23 print("\n=== FILTER EVENS ===")
24 all_nums = [15, 22, 8, 35, 42, 17, 30]
25 evens = []
26
27 for num in all_nums:
28     if _____: # Check if even
29         evens._____(_____)
30
31 print("Original:", all_nums)
32 print("Evens only:", evens)
33
34 # Challenge: Build list of passing grades
35 grades = [45, 72, 68, 55, 82, 90, 38, 75]
36 passing = []
37 # Your code here: add all grades >= 60 to passing list
38
39 print("\nAll grades:", grades)
40 print("Passing grades:", passing)
41 print(f"Pass rate: {len(passing)}/{len(grades)}")

```

✔ Checkpoint: Can you build lists dynamically? Excellent!

6 Exercise 6: Common List Patterns (10 minutes)

Master the patterns every programmer uses!

Open [week7_ex6.py](#) and complete:

```

1 # Pattern 1: Accumulator (sum)
2 numbers = [10, 25, 30, 15, 20]
3 total = 0 # Start at 0
4
5 for num in numbers:
6     total = total + num # Accumulate
7
8 print(f"Sum: {total}")
9
10 # Pattern 2: Find pattern
11 print("\n=== FIND FIRST A GRADE ===")
12 grades = [75, 82, 91, 78, 95, 88]
13 found = False
14
15 for grade in grades:
16     if grade >= ___: # A grade is 90+
17         print(f"Found A grade: {grade}")
18         found = True
19         break # Stop after finding first
20
21 if not found:
22     print("No A grades found")
23
24 # Pattern 3: MinMax pattern

```

```

25 print("\n=== TEMPERATURE TRACKER ===")
26 temperatures = [28, 32, 29, 35, 31, 33, 30]
27
28 # Find hottest day
29 hottest = temperatures[0] # Start with first
30 for temp in temperatures:
31     if temp > hottest:
32         hottest = temp
33
34 print(f"Hottest temperature: {hottest}\textdegree{C}")
35
36 # Find coldest day with position
37 coldest_day = 0 # Index of coldest
38 for i in range(len(temperatures)):
39     if temperatures[i] < temperatures[coldest_day]:
40         coldest_day = i
41
42 print(f"Coldest was day {coldest_day + 1}: {temperatures[coldest_day]}\
    \textdegree{C}")
43
44 # Pattern 4: Count pattern
45 print("\n=== SCORE ANALYZER ===")
46 scores = [85, 92, 78, 95, 88, 72, 90, 85]
47
48 # Count high scores (85+)
49 high_count = 0
50 for score in scores:
51     if -----: # Fill in condition
52         high_count += 1
53
54 print(f"Scores 85+: {high_count} out of {len(scores)}")
55 print(f"Percentage: {high_count/len(scores)*100:.1f}%")
56
57 # Challenge: Find average of top 3 scores
58 # Hint: Use sort() or find max 3 times

```

✔ **Checkpoint:** Patterns mastered? You're becoming a list expert!

7 Lab Summary

✔ **Checkpoint:** Final checkpoint - make sure you have:

- ☐ Created and indexed lists correctly
- ☐ Used `append()`, `insert()`, `remove()`, and `pop()`
- ☐ Searched lists with `in`, `index()`, and `count()`
- ☐ Built lists dynamically with loops
- ☐ Applied accumulator, find, and minmax patterns
- ☐ Solved problems with filtering and counting
- ☐ Tested all your code with different inputs

☐ Worked effectively with your partner

Reflection (2 minutes)

Rate your understanding of today's concepts:

Concept	☹️ Need Help	😊 Getting It	😄 Got It!
List indexing and modification	☐	☐	☐
List methods (append, remove, etc)	☐	☐	☐
Building lists with loops	☐	☐	☐
List patterns (accumulator, find, etc)	☐	☐	☐

Which list method will you use most often?

What's the difference between `remove()` and `pop()`?

😄 Fantastic! You've mastered the power of lists!